FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Advanced)-2021

PAPER –1

TEST DATE: 23-12-2020

ANSWERS, HINTS & SOLUTIONS

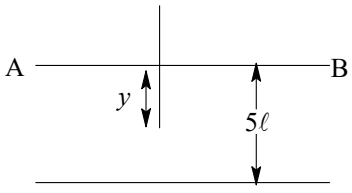
Physics

PART – I

SECTION – A

1. C

Sol. $a=\dfrac{\frac{m}{\ell}yg}{m}=\dfrac{gy}{\ell}$



$$\int_0^u v\,dv=\frac{g}{\ell}\int_0^{\ell} y\,dy$$

$$u=\sqrt{g\ell}.....(1)$$

$$v^2=u^2+2g(4\ell)$$

$$u=3\sqrt{gl}$$

2. A

Sol. $v=\dfrac{C}{\lambda}=RC\left[\dfrac{1}{(n-m)^2}-\dfrac{1}{n^2}\right]=\dfrac{RC}{n^2}\left[\left(1-\dfrac{m}{n}\right)^{-2}-1\right]=\left(\dfrac{2RC}{n^3}\right)m$

For m = 1, 2, 3

v is in A.P.

3. D

Sol. $-(0.7)(10)\left(\frac{1}{2}\right)=\frac{1}{2}(1)v^2-\frac{1}{2}(1)4^2$

$V=3m/s$

$-(0.7)(10)\left(\frac{1}{2}\right)=\frac{1}{2}u^2-\frac{1}{2}3^2$

$v^2=2$

$\Rightarrow -(0.7)(10)x'=0-\frac{1}{2}\times 2,\quad x'=\frac{1}{7}$

So final separation $=2\left(\frac{1}{2}-\frac{1}{7}\right)=\frac{5}{7}m$

4. B

Sol. Work function $=\frac{hC}{\lambda}-\frac{hC}{2\lambda}=\frac{hC}{2\lambda}$

5. C

Sol. $PV^x=\frac{P}{8}(4V)^x$

$8=(4)^x$

$3\ln 2=x2\ln 2$

$x=\frac{3}{2}$

$C=\frac{R}{\frac{5}{3}-1}+\frac{R}{1-\frac{3}{2}}=\frac{3R}{2}-2R=-\frac{R}{2}$

6. D

Sol. $v_1+2v_2=10$

$\frac{v_1-v_2=-e(10)}{3v_2=10(1+e)}$

$3\sqrt{5(10)\left(\frac{1}{2}\right)}=10(1+e)\Rightarrow e=\frac{1}{2}$

7. ABC

Sol. $\Delta w=q\Delta V$

8. BC

Sol. $\frac{N}{r^2}=\text{constant}$

$\frac{N_0}{r^2}=\frac{N}{r^2},\quad N=\frac{N_0}{4}$

Also $\frac{N_0 e^{-\lambda t}}{r^2} = \text{constant}$

9. AD

Sol. $2\left(C_V\right)_{\text{monoatomic}} > \left(C_V\right)_{\text{diatomic}}$

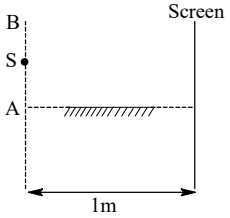
Maximum change $= \frac{(2)\left(\frac{3R}{2}\right) - (1)\left(\frac{5R}{2}\right)}{\frac{5R}{2}} = \frac{1}{5}$

$\%\,\text{change} = 20\%$

10. BC

11. BC

Sol. Let $AS = h$



$\frac{1}{4} = \frac{\lambda(1)}{2h} \ldots\ldots (i)$

$\frac{1}{6} = \frac{\lambda(1)}{2(h+0.6)} \ldots\ldots (ii)$

$\frac{\frac{1}{4}}{\frac{1}{6}} = \frac{h+0.6}{h}$

$\frac{3}{2} = \frac{h+0.6}{h} \Rightarrow h = 1.2mm$

$\frac{1}{4}mm = \frac{\lambda(1m)}{2\times(1.2)mm} \Rightarrow \lambda = 6000\,\text{Å}$

12. AD

SECTION – C

13. 00000.40

Sol. Apply COM

14. 00002.00

Sol. $U.\rho.=\text{Constant}$

$\frac{T}{V}=\text{constant}$

P=constant

$W=nR\Delta T \quad .....(i)$

$n(C_V)\Delta T=3$

$n\left(\frac{3R}{2}\right)\Delta T=3 \Rightarrow W=2\,joule$

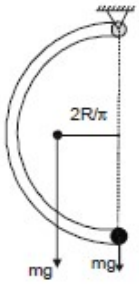
15. 00060.00

16. 00001.50

Sol. $\tau = I_0\alpha$

$mg\frac{2R}{\pi}=\left(2mR^2+4mR^2\right)\alpha$

$\alpha=\frac{mg2R}{\pi 6mR^2}=\frac{g}{3\pi R};\ a_m=2R\times\alpha=\frac{2g}{3\pi}$



17. 00025.00

Sol. $I_1+I_2=13I_0$

$\left(\sqrt{I_1}-\sqrt{I_2}\right)^2=I_0$

$I_1+I_2-2\sqrt{I_1}\sqrt{I_2}=I_0$

$2\sqrt{I_1}\sqrt{I_2}=12I_0$

$I_{max}=13I_0+12I_0=25I_0=(5)^2 I_0$

18. 00003.00

Sol. Let P_1 be density after first stroke $VP=(V+\Delta V)P_1=\frac{V}{(V+\Delta V)}P$

After nth stroke

$P_n=\left(\frac{V}{V+\Delta V}\right)^n P,\quad \frac{P_n}{P}=\left(\frac{V}{V+\Delta V}\right)^n$

Pressure $\propto$ density

$\frac{P_n}{P}=\left(\frac{V}{V+\Delta V}\right)^n$

$\frac{1}{1.728}=\left(\frac{V_0}{1.2V_0}\right)^n$

$(1.2)^n=(1.2)^3$

$n=3$

Chemistry

PART – II

SECTION – A

19. C

Sol. (I) is trans and (II) is cis form

20. A

Sol. $\underbrace{2KI}_{\text{Four ions}} + HgI_2 \longrightarrow \underbrace{K_2[HgI_4]}_{\text{three ions}}$

21. A

Sol. Due to back bonding (A) is a weaker base.

22. C

Sol. Fully filled d-subshell is colourless in nature.

23. D

Sol. $\Delta S = nRln\frac{V_2}{V_1}$

$\Delta H = 0$

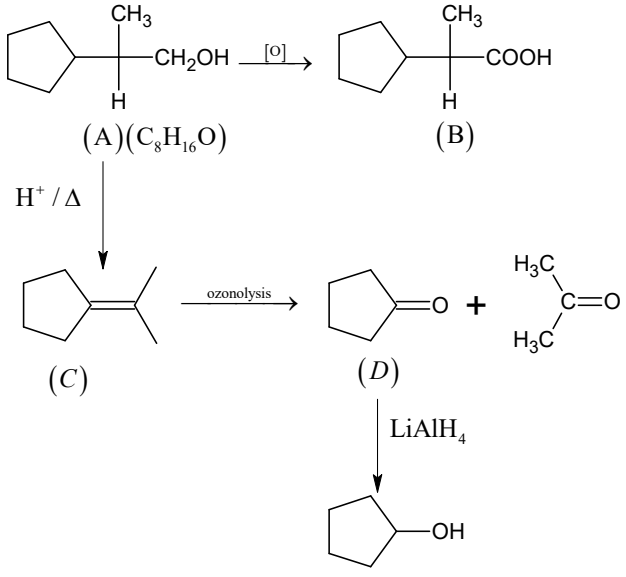
$\Delta G = \Delta H - T\Delta S$

24. C

25. ABCD

26. ABCD

Sol.



27. BC

Sol. A lone pair of N-atom participating in resonance will be less basic

28. AB

Sol. Reaction proceeds by limiting reagent only. Enthalpy of neutralization decreases, when acid or base or both are weak:

29. AD

Sol. BaO_2 and Na_2O_2 are peroxides

30. ABCD

SECTION – C

31. 00252.00

Sol. $$\frac{r_1}{r_2}=\sqrt{\frac{M_2}{M_1}}\times\frac{P_1}{P_2}$$

Or $$\frac{n_1}{t_1}\times\frac{t_2}{n_2}=\sqrt{\frac{M_2}{M_1}}\times\frac{P_1}{P_2} \quad \text{Or} \quad \frac{1}{38}\times\frac{57}{1}=\sqrt{\frac{M}{28}}\times\frac{0.8}{1.6}$$

$\therefore\ M=252$

32. 00167.20

Sol. Given,

$$\frac{1}{2}H_2(g)+(Aq)\rightarrow H^+(aq); \qquad \Delta H^\circ=0.....(\text{i})$$

$$\frac{1}{2}H_2(g)+\frac{1}{2}Cl_2(g)\rightarrow HCl(g); \qquad \Delta H^\circ=-92.4kJ.....(\text{ii})$$

$$HCl(g)+H_2O(l)\rightarrow H_3O^+(aq)+Cl^-(aq); \qquad \Delta H=-74.8kJ.....(\text{iii})$$

By inspection method: $\text{Eqs.}(\text{ii})+(\text{iii})-(\text{i})$ reveals that

$$\frac{1}{2}Cl_2(g)+(Aq)\rightarrow Cl^-(aq); \qquad \Delta H=-167.2kJ$$

i.e. Heat of formation of $Cl^-(aq)=-167.2kJ$

33. 00009.17

Sol. Let solubility of $PbBr_2$ is $S\ mol\ litre^{-1}$

$$PbBr_2(s)+aq\rightleftharpoons PbBr_2(aq)\rightleftharpoons \underset{\frac{S\times80}{100}}{Pb^{2+}}+\underset{\frac{2S\times80}{100}}{2Br^-}$$

Since, $PbBr_2$ ionises to 80% only.

Now $$K_{sp}=\left[Pb^{2+}\right]\left[Br^-\right]^2$$

Or $$3.2\times10^{-5}=\left[\frac{S\times80}{100}\right]\left[\frac{2S\times80}{100}\right]^2$$

$$S=0.025\,mol\,litre^{-1}$$

$$=0.025\times367\,g\,litre^{-1}$$

$$=9.175\,g\,litre^{-1}$$

34. 00098.00

Sol. $T = 20°C$ $0.1m$ solution of urea ΔP=0.03mm

$KCl \to 0.1m \quad \Delta P = 0.0594$

$$\frac{\Delta P}{P} = x_\beta \to 0.1 mole in 1kg$$

$$\frac{0.03}{P} = \frac{0.1}{0.1+55.5}$$

$$P = \frac{0.03 \times 55.65}{0.1} = 16.695$$

$$\frac{\Delta P}{P} = iX_\beta$$

$$\frac{0.0594}{16.695} = i\left[0.00179\right]$$

$$i = 1.987$$

35. 00036.00

36. 00030.60

Sol. $$KE = 13.6 \times \frac{Z^2}{n^2}$$

$$= 13.6 \times \frac{3^2}{2^2} = 30.6\text{eV}$$

Mathematics

PART – III

SECTION – A

37. B

38. A

Sol. As s–b, s–c, s–a are in AP with common difference =1

$\Rightarrow$ a,c, b are in AP with common difference equal to 1.

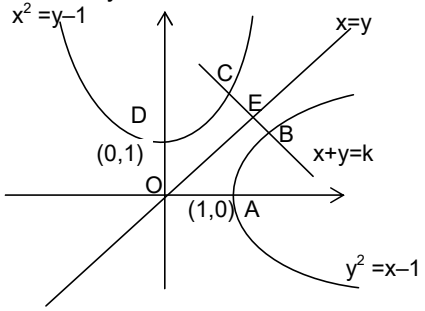
$\Rightarrow$ c = a+ 1, b = a+2 $\quad \Rightarrow s = \dfrac{a+a+1+a+2}{2} = \dfrac{3(a+1)}{2} \Rightarrow \Delta = \sqrt{\dfrac{3}{2}(a+1).\dfrac{a+3}{2}.\dfrac{a+1}{2}.\dfrac{a-1}{2}}$

Also r = Δ/s $\Rightarrow a^2 + 2a - 195 = 0 \Rightarrow$ (a+15)(a–13)=0

A = 13, c=14, b=15 $\quad \Rightarrow \Delta$ = 84 sq. units

39. C

Sol. Line of shortest distance will be perpendicular to line x=y. Also both parabolas are symmetric about x=y.



A=area OABCDO

A′=area OECDO

$\Rightarrow$ A′ =A/2

40. C

Sol. Let α be the real root then $(a\alpha^2 + 2b\alpha + c) + i(d\alpha^2 + 2e\alpha + f) = 0 \quad \Rightarrow a\alpha^2 + 2b\alpha + c = 0 \Rightarrow \alpha = \dfrac{-\sqrt{c}}{\sqrt{a}}$

$\Rightarrow d\alpha^2 + 2e\alpha + f = 0 \quad \Rightarrow d.\dfrac{c}{a} - 2e\dfrac{\sqrt{c}}{\sqrt{a}} + f = 0 \quad \Rightarrow \dfrac{d}{a} - 2.\dfrac{e}{b} + \dfrac{f}{c} = 0$

41. B

Sol. $I = \int \tan^{-p/3} x.\sec^4 x dx \quad = \int \tan^{-p/3} x(1+\tan^2 x).\sec^2 x dx$

42. B

Sol. If equation of line AB is $\vec{r} = \vec{0} + \lambda(\sqrt{2}a\hat{j} - \vec{0}) = \lambda\hat{j}$

then equation of line DC is $\vec{r} = \sqrt{2}a\hat{i} + \mu\left(\dfrac{a}{\sqrt{2}}\hat{i} + \dfrac{a}{\sqrt{2}}\hat{j} + a\hat{k} - \sqrt{2}a\hat{i}\right)$

$\vec{r} = \sqrt{2}a\hat{i} + \mu\left(\dfrac{-1}{\sqrt{2}}\hat{i} + \dfrac{1}{\sqrt{2}}\hat{j} + \hat{k}\right) \Rightarrow$ S.D $= \left|\dfrac{(\vec{a}_2 - \vec{a}_1).(\vec{b}_1 \times \vec{b}_2)}{\left|\vec{b}_1 \times \vec{b}_2\right|}\right| = \dfrac{2a}{\sqrt{3}}$

43. BCD

Sol. $I = \int_{-1}^{1} \left(e^{x^3} + e^{-x^3} \right) dx > 2$

Graph is concave upwards

$\Rightarrow I < f(0) + f(1)$

44. ACD

Sol. $f'(x) = e^{-\frac{1}{x^2}} . \frac{2}{x^3} + \frac{\pi}{2} \sqrt{1 + \sin \frac{\pi x}{2}}$

$\lim_{x \to 0^+} \frac{e^{-\frac{1}{x^2}}}{x^3} = 0$

And

Also $\lim_{x \to \infty} f(x) = \infty$

f''(x) does not exist for x=3, 7, 11….

45. AD

Sol. $\frac{2x^4 + 2y^4 + 4z^4 + 1}{4} \geq 2xyz$

And $x^4 y + xy^4 + \frac{4}{x^2 y^3} + \frac{1}{x^3 y^2} + 8 \geq 5 - 2$

$\Rightarrow \ell_2 > 10$ (equality cannot hold)

46. AB

Sol. (Put $x = y + \frac{1}{2}$)

47. AB

Sol. $\left(\frac{1}{x-1} + 1 \right) - \left(\frac{4}{x-4} + 1 \right) - \left(\frac{5}{x-5} + 1 \right) + \left(\frac{8}{x-8} + 1 \right) = \frac{6x^2 - 27x}{40}$

$\Rightarrow x = 0$

Also $\frac{2x-9}{(x-1)(x-8)} - \frac{2x-9}{(x-4)(x-5)} = \frac{3}{40}(2x-9)$

And x=9

48. ABD

Sol. Condition is satisfied for (a, b+c-2a)

$\Rightarrow b + c = 2003$

SECTION – C

49. 00000.32

50. 00031.00

Sol. $\dfrac{(x-3)^{\frac{-|x|}{x}}\sqrt{(x-4)^2}\,(17-x)}{\sqrt{-x}\left(-x^2+x-1\right)\left(|x|-32\right)}<0$

For $\sqrt{-x}$ to be defined $x<0$

$\Rightarrow \dfrac{-|x|}{x}=1$

Since $\sqrt{(x-4)^2},\sqrt{-x}$ are positive & $-x^2+x-1$ is always negative

$\dfrac{(x-3)(x-17)}{x+32}>0$

$x<0 \quad \Rightarrow \dfrac{1}{x+32}>0 \Rightarrow x=31$

51. 00833.33

Sol. $\because \int_a^x e^{f(t)}dt=\int_0^x g(x-t)dt+2x+3$

$\Rightarrow \int_a^x e^{f(t)}dt=\int_0^x g(t)dt+2x+3$ (Using King Property)

Differential both sides, we get

$e^{f(x)}=g(x)+2$

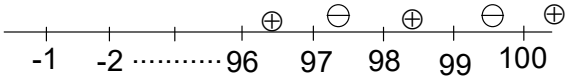
$\Rightarrow g(x)=e^{f(x)}-2$

$\Rightarrow g'(x)=e^{f(x)}.f'(x)$

$\because e^{f(x)}$ is always greater than zero.

$\therefore$ sign of g'(x) is same as sign of $f'(x)$.

$\therefore$ sign of g'(x)



Clearly, local extremum (maximum or minimum) will occur at x=99,97,95,….,3,1

$\therefore$ Sum of all the values=1+3+5+…+99= $\dfrac{50}{2}[2\times1+(50-1)\times2]=2500$

52. 00008.00

Sol. E_1 : Dot removed from odd face.

$P(E_1)=\dfrac{9}{21}$

E_2 : Dot removed from even face.

$P(E_2)=\dfrac{12}{21}$ E: Die shows odd numbers of dots.

$P(E)=P(E\cap E_1)+P(E\cap E_2)=\dfrac{11}{21}$

53. 00009.00

Sol. $\left|Z\left(a+bi\right)-Z\right|=\left|(a+bi)Z\right|$

$\Rightarrow a=\frac{1}{2}$

$a^2+b^2=64$

54. 00002.00

Sol. $\begin{bmatrix} 1 & 2n & na+8\sum_{k=0}^{n-1}k \\ 0 & 1 & 4n \\ 0 & 0 & 1 \end{bmatrix}$